A finite check of the Murty–Simon lemmas of Das–Fox on all diameter-2-critical graphs with at most 11 vertices

(a sanity check of claims by others; no novelty is claimed)

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-11-01

doi:10.5281/zenodo.23864744 (not peer reviewed)

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** All eight corrections are applied (Section 4). No human audit exists. There is no Lean in this paper.

Abstract

Das and Fox (arXiv:2610.39032, 28 Oct 2026) claim a proof of the Murty–Simon conjecture: a diameter-2-critical (D2C) graph on n vertices has at most $\lfloor n^2/4 \rfloor$ edges. Two of their lemmas are unconditional statements about all D2C graphs: Lemma 1.2 (for every maximal matching M, $|E \setminus M|$ is at most the number of non-edges) and Lemma 1.3 (if equality holds for every maximal matching, the graph is triangle-free). **The proof is theirs; we claim nothing new and do not verify it.** We test the two lemma statements, and Theorem 1.1, on every D2C graph with $3 \le n \le 11$: 4,419 graphs and 776,217 maximal matchings, with exact integer arithmetic and two independent implementations (a C filter with Python, and a Rust filter with a full fresh enumeration of all 1,006,700,565 connected graphs on 11 vertices). There are no violations. The D2C counts equal OEIS A330333, so the census is a recomputation; Theorem 1.1 was already reported proved for $n \le 24$ and $n = 26$ (Fan 1987, as reported by Dailly–Foucaud–Hansberg), so this is a sanity check, not new evidence. Tier [C] throughout.

What this is NOT. Not a verification of the Das–Fox proof; not evidence for any $n \ge 12$; not a new enumeration or new count; not a formal proof (no Lean); the intermediate lemmas (2.4, 2.5, 2.13, 3.1, 3.2) were checked only by the producer; not human-audited.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code (integers) with the argument written down, not formalised. [X]: unaudited or non-exhaustive items. Identifiers `MS-·-nnnn` refer to `ledger/ledger.json`.

1 Statements tested

A graph is *diameter-2-critical* (D2C) if it has diameter exactly 2 and deleting any edge destroys this (K_n is excluded). A *maximal matching* is a matching to which no edge can be added. A *non-edge* is a pair of distinct non-adjacent vertices. As read by the auditor from the full text of Das–Fox [1] ([L], `MS-L-0001`):

Lemma 1.2. For every D2C graph G on $n \ge 3$ vertices and every maximal matching M: $|E(G) \setminus M| \le |\overline{E}(G)|$.

Lemma 1.3. If equality holds for *every* maximal matching M, then G is triangle-free.

Theorem 1.1. A D2C graph on n vertices has at most $\lfloor n^2/4 \rfloor$ edges.

Both lemmas carry no extra hypothesis, so they are directly testable on a finite range. Attribution, from the paper's AI statement: GPT-5.6 Sol found a proof of the $\lfloor n^2/4 \rfloor$ bound (Lemma 1.2) and an equality-case proof for even n only; the equality-case proof for all n (Lemma 1.3) is the authors' own.

2 Census and lemma test, $3 \le n \le 11$ ([C])

Producer: `nauty-geng -c` piped into a C filter (`d2c_filter.c`), then Python (`lemmas.py`) for maximal matchings and lemma checks. Auditor, without running producer code: fresh `geng`, a Rust filter using a common-neighbour characterisation of edge criticality (cross-checked by explicit BFS edge deletion for $n \le 10$, 0 disagreements on 11,989,762 connected graphs), a full fresh run over all 1,006,700,565 connected graphs at $n = 11$, and its own maximal-matching enumerator (equal to brute force over all edge subsets on the 51 D2C graphs with $n \le 8$ plus 14 control graphs). Canonical forms (`labelg`) of the two D2C sets are equal for every $n = 3, \ldots, 11$: symmetric difference 0 (`MS-C-0001`, `-0002`).

n	connected	D2C	max edges	$\lfloor n^2/4 \rfloor$	max. matchings	violations
3	2	1	2	2	2	0
4	6	2	4	4	5	0
5	21	3	6	6	15	0
6	112	5	9	9	39	0
7	853	10	12	12	147	0
8	11,117	30	16	16	788	0
9	261,080	103	20	20	5,142	0
10	11,716,571	519	25	25	50,991	0
11	1,006,700,565	3,746	30	30	719,088	0
total		4,419			776,217	0

Table 1: Connected graphs equal OEIS A001349. "Violations" counts violations of Lemma 1.2 and of Lemma 1.3 (both 0 for every n). At every n the maximum number of edges is attained uniquely by the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$.

Lemma 1.3 is nearly vacuous here. Exactly one graph per n is an equality case for *all* maximal matchings: the balanced $K_{a,b}$, which is triangle-free. The auditor's additional finding: equality for even *one* maximal matching occurs only at $K_{a,b}$. D2C graphs containing a triangle ($1, 4, 20, 87, 488, 3685$ graphs for $n = 6, \ldots, 11$) stay strictly below the bound of Lemma 1.2, by at least 1 at $n = 6$ and at least 10 at $n = 11$. Lemma 1.2 on 776,217 maximal matchings is the informative test.

Controls. Positive: C_4, C_5, Petersen, $K_{2,3}$, $K_{3,3}$, $K_{4,5}$, $K_{1,5}$ accepted by both D2C filters; negative: K_3, K_4, K_5, $K_4 - e$, C_6, C_7, P_4 rejected. Connected-graph counts equal A001349; networkx's graph atlas reproduces the D2C counts for $n \le 7$. Perturbed statements that must fail do fail: strict "$<$" in Lemma 1.2, the bound minus one, and the empty (non-maximal) matching all fail at $K_{a,b}$.

Structural facts of Dailly–Foucaud–Hansberg. On the auditor's D2C list we reproduce: 13 non-bipartite D2C graphs at $\lfloor (n-1)^2/4 \rfloor + 1$ edges outside the C_5^+ family (orders $7, 8, 9$: $3, 8, 2$ graphs; 3 without a dominating edge); 10 graphs at that bound with a dominating edge, all with $n \le 9$; only H_5 ($n = 6$, 8 edges) above that bound; maximum degree $n - 1$ only in stars ([C], `MS-C-0003`).

Intermediate lemmas ([X]). Lemmas 2.4, 2.5, 2.13, 3.1, 3.2 of the paper, as implemented by the producer from the paper's definitions (critical non-edge, I, I', C_ℓ), have 0 violations on all 4,419 graphs. The auditor did not recompute them (`MS-X-0001`).

3 Prior art and strength of the evidence

The D2C counts $1, 2, 3, 5, 10, 30, 103, 519, 3746$ are OEIS A330333 (Zivkovic [3], Dec 2019; $a(11)$ added by S. A. Irvine, 5 Sep 2026): `https://oeis.org/A330333` ([L], `MS-L-0002`). Dailly–Foucaud–Hansberg [2] report a computer search over all graphs of order ≤ 11 but publish no per-order counts; MacDougall–Eggleton (1997) list the D2C graphs up to $n = 7$ (via the auditor). The census is therefore a recomputation. Fan (1987) proved

the Murty–Simon conjecture for $n \leq 24$ and $n = 26$ (as reported by [2]; Fan was not read), so our check of Theorem 1.1 re-confirms an established range. The only possibly new element is the per-graph test of Lemmas 1.2–1.3; we did not search whether anyone has run it, and it is weak evidence about the proof.

4 Audit corrections applied

The producer's `data/RESULT.md`, `data/PREREG_ITEM.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them as follows (a dated erratum, not a silent edit). (1) The counts equal OEIS A330333; the census is a recomputation. (2) DFH give no per-order counts; MacDougall–Eggleton list $n \leq 7$. (3) Fan (1987) covers $n \leq 24$, $n = 26$ (via DFH); our Theorem 1.1 check is a sanity check. (4) "Written before any computation" is not verifiable: `PREREG_ITEM.md` (sha256 `b731521c...7d56`) and the results were committed together (`c981c12`); self-attested, hash recorded. (5) The DFH structural facts, listed as not tested, were reproduced by the auditor. (6) Process count: 4 `geng`|filter pipelines = 8 processes on cores 2–5, not ≤ 4 (the auditor's $n = 11$ run used the same layout). (7) AI attribution as in Section 1. (8) The slack finding of Section 2.

5 Limitations and what is not done

- AI audit only; no human audit; no Lean.
- Not verified: the logic of the Das–Fox proof; intermediate Lemmas 2.4, 2.5, 2.13, 3.1, 3.2 (producer only); anything for $n \geq 12$ ($n = 12$ has about 1.65×10^{11} connected graphs, about 16 wall-hours on 4 cores; not run); Fan (1987); the producer's reconstructed Figure 1 graph.
- The readings of "maximal matching" and "diameter-2-critical" follow the paper's text as read by the auditor.
- Trusted components: `nauty` 2.7 (`geng`, `labelg`), a C compiler, `rustc`, Python integers.

6 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code, per-n D2C graph6 lists (`.g6.txt`), lemma summaries and logs are in `data/`; the auditor's Rust source, Python scripts and small outputs are in `data/audit` (the full 10^9-graph enumeration outputs and binaries are not copied). The paper's full text is third-party and is not shipped. Nothing was committed, pushed or uploaded.

References

[1] S. Das, J. Fox, *A proof of the Murty–Simon conjecture on diameter-2-critical graphs*, arXiv:2610.39032 (v1, 28 Oct 2026), unrefereed. ([L])

[2] A. Dailly, F. Foucaud, A. Hansberg, *Strengthening the Murty–Simon conjecture on diameter 2 critical graphs*, arXiv:1812.08420 (Dec 2018); Discrete Math. 2019. ([L])

[3] M. Zivkovic, OEIS A330333, *Number of diameter-2-critical graphs with n nodes*, `https://oeis.org/A330333`. ([L])